

RENEWABLE RESOURCES

A renewable resource is a natural resource that can be re-made or re-grown at a scale comparable to its consumption.

GLOBAL ENERGY STATISTICAL YEARBOOK 2016

Share of renewable in electricity Production (incl. hydro) Norway (97.9%)

Global Extreme in 2

Share of wind and solar in electricity Production Portugal (24.2%)

What renewables are

❑ ***Renewable energy*** – "any sustainable energy source that comes from the natural environment."

❑ Some features of Renewable energy

- ✓ It exists perpetually and in abundance in the environment
- ✓ ready to be harnessed, inexhaustible
- ✓ It is a clean alternative to fossil fuels

"energy that is derived from natural processes that are replenished constantly" – defined by the International Energy Agency

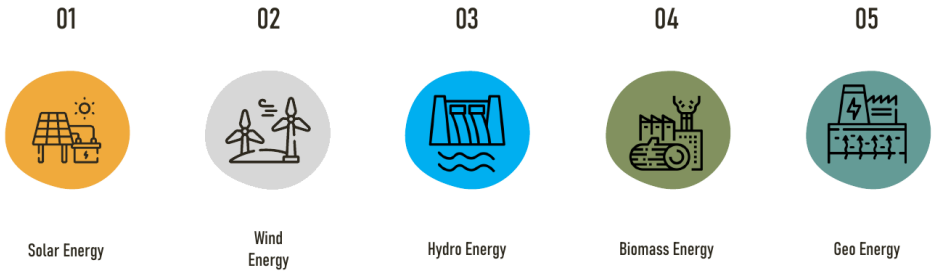
Major Renewable Energy Sources

- ❑ Hydro energy
- ❑ Wind energy
- ❑ Solar energy
- ❑ Biomass energy
- ❑ Tidal energy
- ❑ Geothermal energy
- ❑ Wave energy
- ❑ Bio-fuel
- ❑ Biogases

• These sources are normally used to produce **clean** (or **green**) energy. This production does not lead to climate change and does not involve the emission of pollutants.

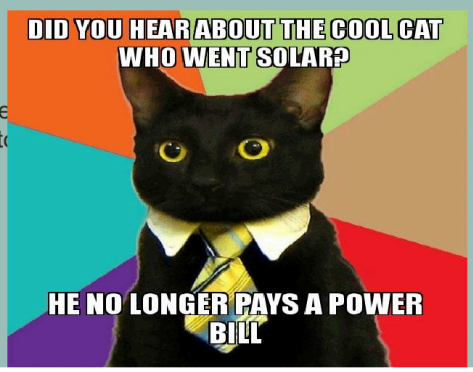
• A related term is **sustainable** energy: this concept refers to generating energy with an awareness of the future, i.e., in a way that would enable future generations to meet their energy needs too. The concept is related not only to renewable energy but also to **energy efficiency**.

TYPES OF RENEWABLE ENERGY



SOLAR ENERGY

Solar power is the curre conversion of energy from photo sunlight into electricity, directly using photovoltaic cells, indirectly using concentrated solar power, or a combination. Photovoltaic cells convert light into an electric



Benefits of Solar Energy:-

Impact on the Environment

Reduce Your Energy Bill

Solar Energy Is Applicable Everywhere

Less Electricity Loss During Long-Distance Transport

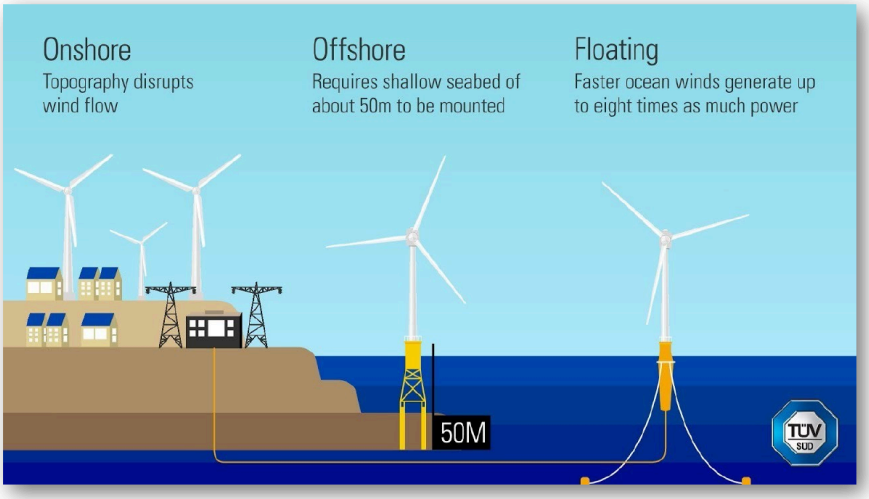
WIND ENERGY

Wind power or wind energy describes the process by which the wind is used to generate mechanical power or electricity.

They are usually grouped in wind farms (sometimes called wind parks). There are onshore farms (which, however, are often near water); near-shore farms (on land or sea within several km of the coast); and offshore parks



Tauern wind park, Austria



Perks of hydro energy-

1. The cheapest energy source
Even though the initial investment required to build a plant is very high and something of a challenge, waterpower is the cheapest source overall in the medium to long term. Once the dams and stations have been built and the turbines installed, they require only minimally expensive maintenance compared to the initial investment.

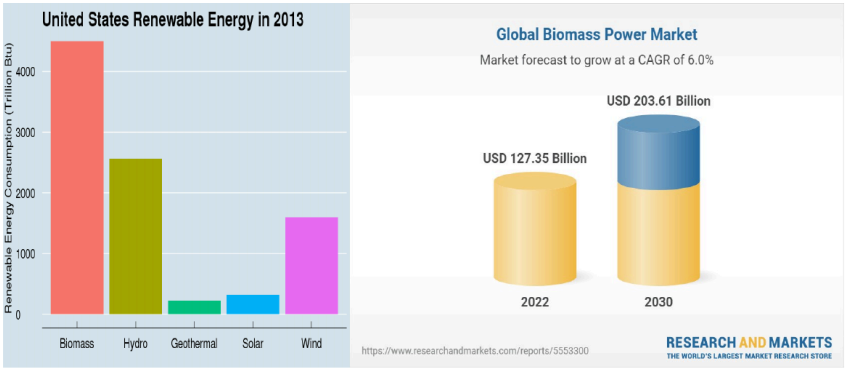
2. Hydropower stations are agile and responsive
Faced with water availability that can change over time, the power stations are extremely flexible. Production systems require a very small amount of energy to start working.

3. Water brings with it enormous energy
The energy potential of hydropower is enormous. The gigantic masses of water found at high altitudes have a significant amount of gravitational potential energy, and even using just a part of it will provide abundant power.

4. Production can be tailored to demand
The water flow can be easily controlled to suit energy requirements. This is true both in the short term – for example, in the day-night cycle – and in the longer term for seasonal variations and periods of drought.

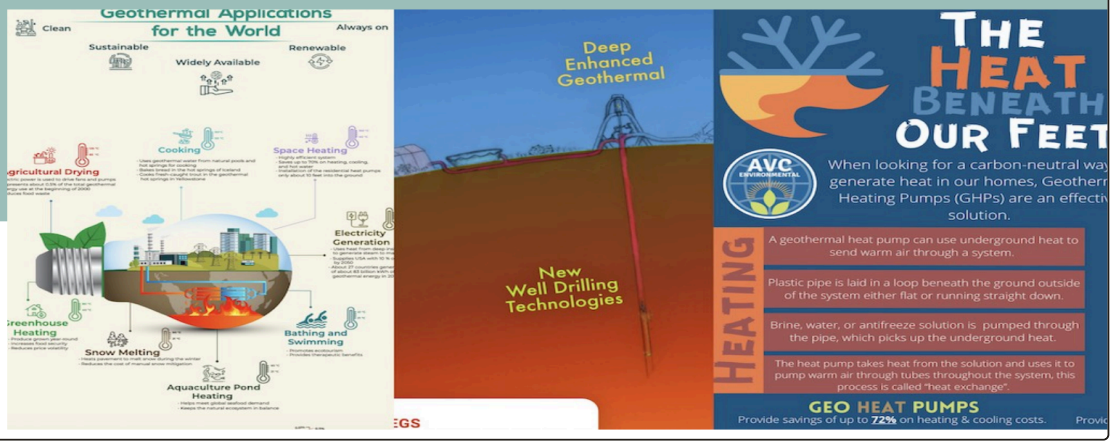
5. Hydropower as a tool for reclamation...
With water courses contained within pre-established reservoirs, hydropower stations also help avoid flooding, to drain marshy land where stagnant water tends to accumulate. The flow and total water volume can be controlled very precisely. Thus, it is possible to ensure that the water flows steadily, even where rainfall tends to be intense and concentrated in short periods.

6. ... and as an irrigation system
At the opposite end of the spectrum to the previous benefit, hydropower stations can also prove invaluable where water is very scarce. Reservoirs can be used as water reserves for periods of drought. Usually, the water coming out of the station is perfectly clean and usable. The opening of dams can also be planned and calibrated around crop cycles and ecosystems, and the water can be allowed to flow to allow fish to get through at appropriate times.



GEOTHERMAL ENERGY

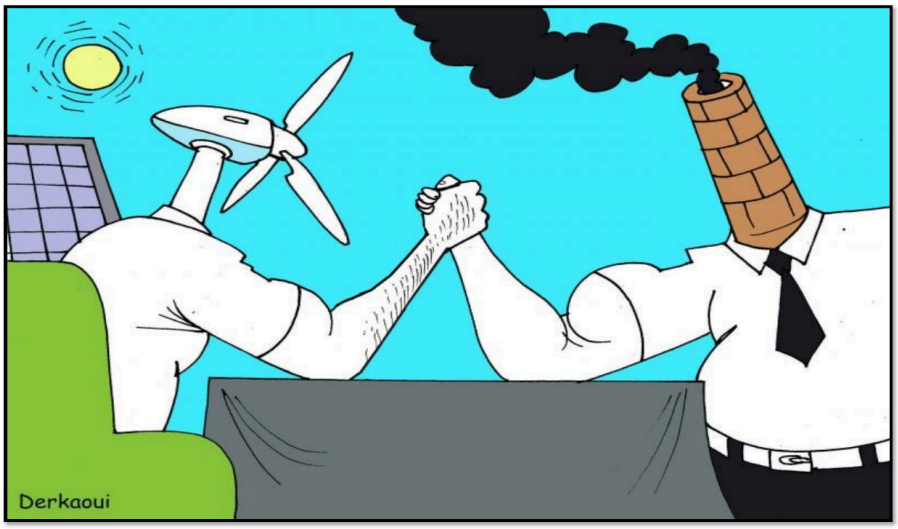
Geothermal energy is heat that is generated within the Earth. (Geo means “earth,” and thermal means “heat” in Greek.) It is a renewable resource that can be harvested for human use. About 2,900 kilometers (1,800 miles) below the Earth's crust, or surface, is the hottest part of our planet: the core.



Most small island economies rely on oil-fired power plants to provide a steady electricity supply, but **Iceland** has virtually 100% renewable electricity from its abundant hydropower and geothermal resources.



WHO WILL WIN THIS BATTLE ?

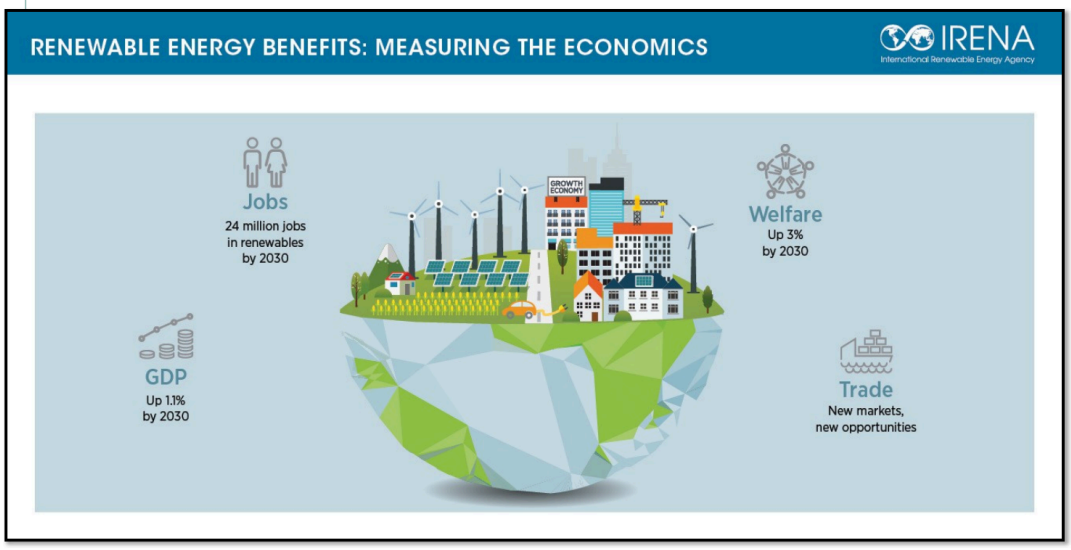


BENEFITS OF RENEWABLE ENERGY

- ❑ 46M pounds of annual reductions in GHG emissions.
- ❑ 10M gallons in annual water annually savings
- ❑ 225 acres of land preserved by using roofs and parking lots



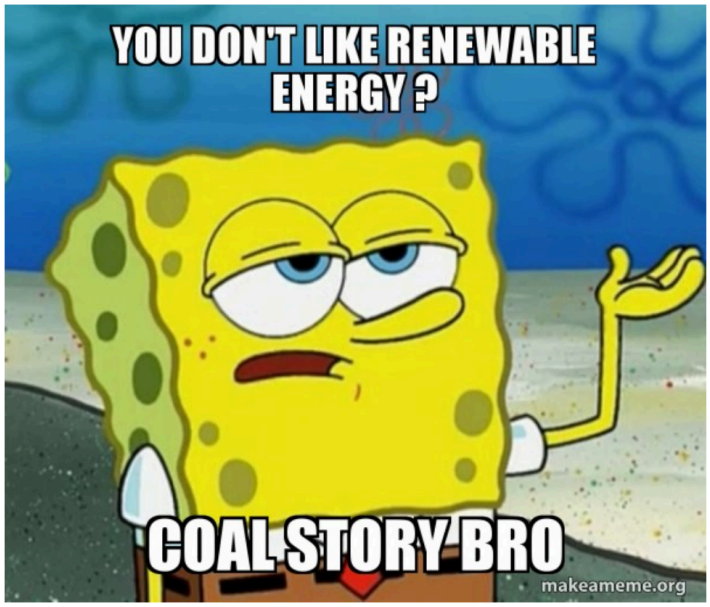
BENEFITS OF RENEWABLE ENERGY



Feed-in tariffs

Since renewables are still innovative and in active development, they are often not competitive with traditional energy sources. Therefore, green-minded governments provide various incentives that encourage investments in the sector and promote its faster development. Among the most common is the feed-in tariff. This is an obligation imposed on utilities by the government **to buy a certain amount of renewable electricity at prices higher than the market's rates.** The higher expenditure for the utility is passed on to its customers. The increase in prices that customers must bear is usually small, but these small contributions are a powerful and effective way to support green energy.

Feed-in tariffs were introduced as early as 1978 in the USA. Now, they are implemented in around 50 countries around the world. In Germany, for example, feed-in tariffs are regulated by the Renewable Energy Law (Erneuerbare-Energien-Gesetz). The programme adds around EUR 1 to each monthly residential electricity bill, which translates into billions of euros of subsidies for the energy sector each year. The country aimed to generate 12.5% of its electricity from renewable sources by 2010. The percentage should rise to 20 by 2020.



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